



Procesamiento de datos {dplyr}

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Procesamiento, visualización y modelado de datos en R



clase_r

⊕ New ▾ | 📁 Open ▾ | 💾 | 📄 | 🔍 Search ▾ | R 4.5.1 | 📁 clase_r ▾

CONSOLE | TERMINAL | PROBLEMS | OUTPUT | ... | + ▾ | - | × | 📄 | 📄 | ☰

R 4.5.1 started.

R version 4.5.1 (2025-06-13) -- "Great Square Root"
Copyright (C) 2025 The R Foundation for Statistical Computing
Platform: aarch64-apple-darwin20

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>

Abra Positron y verifique que está dentro de su folder de trabajo **clase_r**

Si no está dentro de su folder entonces haga clic en la barra de menú **File/Open Folder...**



Su turno...

- Genere un nuevo script
- Agregue encabezado
- Active en su sesión la librería tidyverse

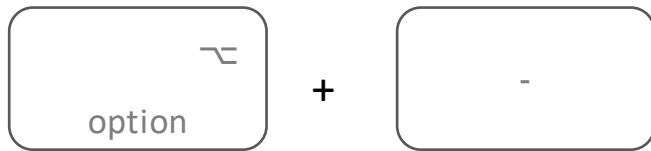
```
code > S05_Tex.R
1 # ===== #
2 # Script de la Sesión 05
3 # Autor: Tex
4 # Curso: "Procesamiento y visualización de datos espaciales en R"
5 # Fecha: 02 de octubre de 2025
6 # ===== #
7
8 # Activo paquetes a utilizar en mi sesión
9 library(tidyverse)
10
```

Ln 14, Col 1 Spaces: 2 UTF-8 LF {} R

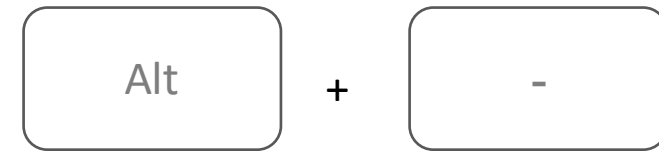


Insertar operador de asignación

<-

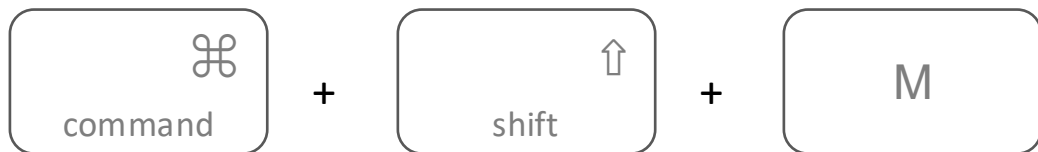


Mac

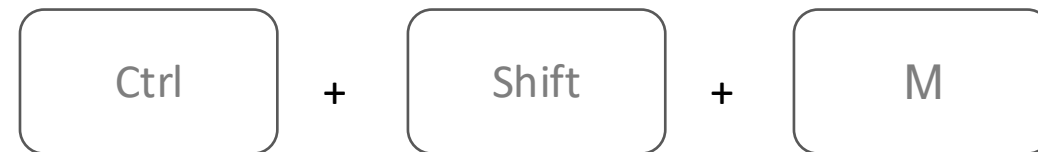


Windows

Insertar pipe

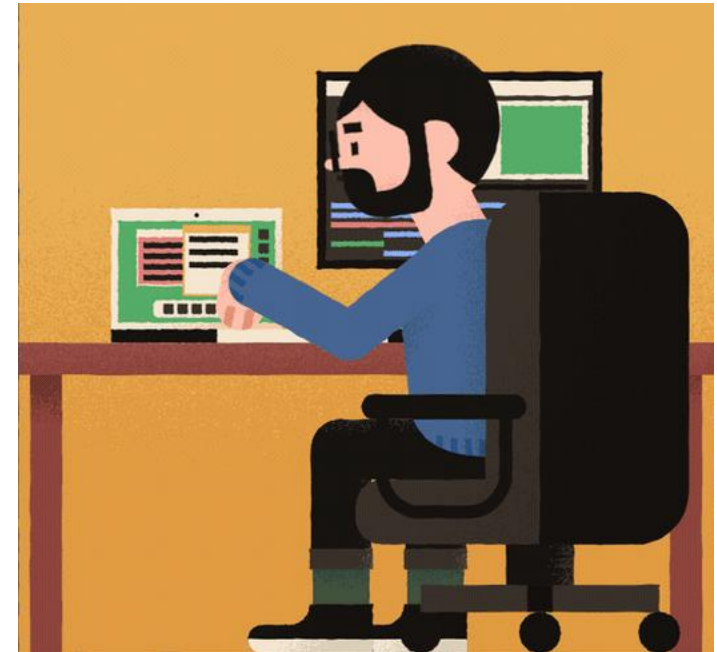


Mac

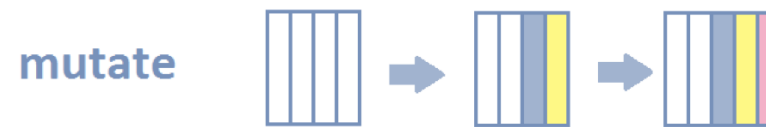
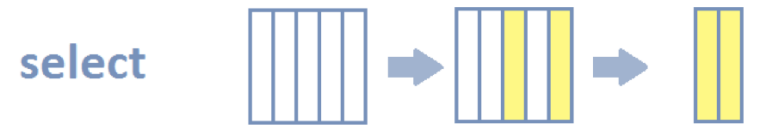


Windows

Procesamiento de datos

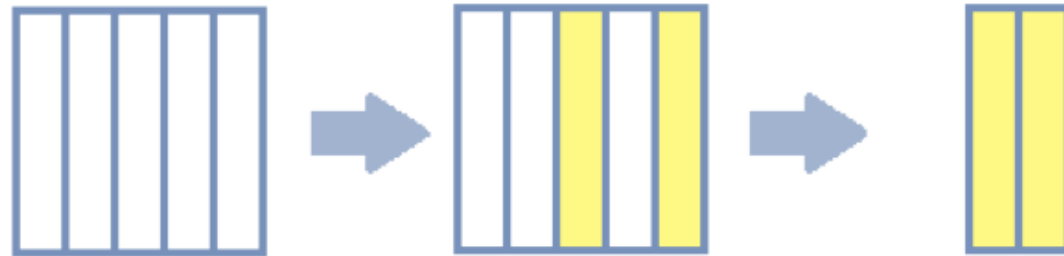


{dplyr} manipulación datos



select()

select





Generamos malla de trabajo para la sesión

```
aire_tbl <- as_tibble(airquality) %>% print()
```

```
# A tibble: 153 × 6
```

	Ozone	Solar.R	Wind	Temp	Month	Day
	<int>	<int>	<dbl>	<int>	<int>	<int>
1	41	190	7.4	67	5	1
2	36	118	8	72	5	2
3	12	149	12.6	74	5	3
4	18	313	11.5	62	5	4
5	NA	NA	14.3	56	5	5
6	28	NA	14.9	66	5	6
7	23	299	8.6	65	5	7
8	19	99	13.8	59	5	8
9	8	19	20.1	61	5	9
10	NA	194	8.6	69	5	10

```
# i 143 more rows
```

```
# i Use `print(n = ...)` to see more rows
```

Seleccionar columnas: `select( )`

```
> aire_tbl
# A tibble: 153 × 6
  Ozone Solar.R Wind Temp Month Day
  <int>   <int> <dbl> <int> <int> <int>
1     41     190   7.4     67     5     1
2     36     118    8      72     5     2
3     12     149  12.6     74     5     3
4     18     313  11.5     62     5     4
5    NA      NA  14.3     56     5     5
6     28      NA  14.9     66     5     6
7     23     299   8.6     65     5     7
8     19      99  13.8     59     5     8
9      8      19  20.1     61     5     9
10    NA     194   8.6     69     5    10
# i 143 more rows
# i Use `print(n = ...)` to see more rows
```

mall %>% select(...)

dataset

Argumentos de selección

Seleccionar columnas específicas

```
aire_tbl %>% select(Day, Month, Ozone)
```

```
# A tibble: 153 × 3
```

```
  Day Month Ozone
  <int> <int> <int>
1     1     5    41
2     2     5    36
3     3     5    12
4     4     5    18
5     5     5    NA
6     6     5    28
7     7     5    23
8     8     5    19
9     9     5     8
10    10     5    NA
```

```
# i 143 more rows
```

```
# i Use `print(n = ...)` to see more rows
```

Seleccionar columnas: `select( )`

```
> aire_tbl
# A tibble: 153 × 6
  Ozone Solar.R Wind Temp Month Day
  <int>   <int> <dbl> <int> <int> <int>
1     41     190   7.4     67     5     1
2     36     118    8      72     5     2
3     12     149  12.6     74     5     3
4     18     313  11.5     62     5     4
5    NA      NA  14.3     56     5     5
6     28      NA  14.9     66     5     6
7     23     299   8.6     65     5     7
8     19      99  13.8     59     5     8
9      8      19  20.1     61     5     9
10    NA     194   8.6     69     5    10
# i 143 more rows
# i Use `print(n = ...)` to see more rows
```

mall %>% select(...)

dataset

Argumentos de selección

Seleccionar un rango de columnas

```
aire_tbl %>% select(Temp:Day)
```

```
# A tibble: 153 × 3
  Temp Month Day
  <int> <int> <int>
1     67     5     1
2     72     5     2
3     74     5     3
4     62     5     4
5     56     5     5
6     66     5     6
7     65     5     7
8     59     5     8
9     61     5     9
10    69     5    10
# i 143 more rows
# i Use `print(n = ...)` to see more rows
```

Seleccionar columnas: `select( )`

```
> aire_tbl
```

```
# A tibble: 153 × 6
```

	Ozone	Solar.R	Wind	Temp	Month	Day
	<int>	<int>	<dbl>	<int>	<int>	<int>
1	41	190	7.4	67	5	1
2	36	118	8	72	5	2
3	12	149	12.6	74	5	3
4	18	313	11.5	62	5	4
5	NA	NA	14.3	56	5	5
6	28	NA	14.9	66	5	6
7	23	299	8.6	65	5	7
8	19	99	13.8	59	5	8
9	8	19	20.1	61	5	9
10	NA	194	8.6	69	5	10

```
# i 143 more rows
```

```
# i Use `print(n = ...)` to see more rows
```

mallá %>% `select(...)`

dataset

Argumentos de selección

Seleccionar columnas a excluir

```
aire_tbl %>% select(-c(Temp, Wind))
```

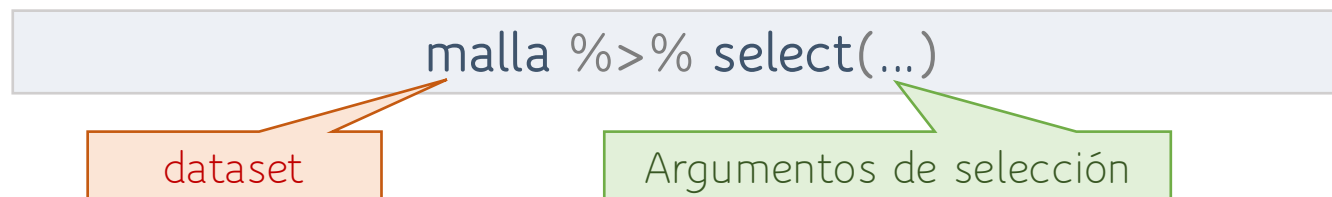
```
# A tibble: 153 × 4
```

	Ozone	Solar.R	Month	Day
	<int>	<int>	<int>	<int>
1	41	190	5	1
2	36	118	5	2
3	12	149	5	3
4	18	313	5	4
5	NA	NA	5	5
6	28	NA	5	6
7	23	299	5	7
8	19	99	5	8
9	8	19	5	9
10	NA	194	5	10

```
# i 143 more rows
```

```
# i Use `print(n = ...)` to see more rows
```

Seleccionar columnas: `select( )`



`starts_with`: Comienzan con este texto

`ends_with`: Terminan con este texto

`contains`: Contienen este texto

`matches`: Coinciden con esta expresión regular

`num_range`: Están en estas posiciones

`one_of`: alguna de estas

`everything`: Todas

filter()

filter



Filtrar filas: `filter( )`

`ma1la %>% filter(...)`

dataset

Argumentos de filtrado

<code>x < y</code>	menor qué
<code>x > y</code>	mayor qué
<code>x == y</code>	igual a
<code>x <= y</code>	menor o igual a
<code>x >= y</code>	mayor o igual a
<code>x != y</code>	diferente de
<code>x %in% y</code>	pertenece a
<code>is.na(x)</code>	is NA
<code>!is.na(x)</code>	Distinto de NA



Filtrar filas: filter()

mallá %>% filter(...)

dataset

Argumentos de filtrado

```
> aire_tbl
```

```
# A tibble: 153 × 6
```

	Ozone	Solar.R	Wind	Temp	Month	Day
	<int>	<int>	<dbl>	<int>	<int>	<int>
1	41	190	7.4	67	5	1
2	36	118	8	72	5	2
3	12	149	12.6	74	5	3
4	18	313	11.5	62	5	4
5	NA	NA	14.3	56	5	5
6	28	NA	14.9	66	5	6
7	23	299	8.6	65	5	7
8	19	99	13.8	59	5	8
9	8	19	20.1	61	5	9
10	NA	194	8.6	69	5	10

```
# i 143 more rows
```

```
# i Use `print(n = ...)` to see more rows
```

Filtro temperatura por arriba de 77

```
aire_tbl %>% filter(Temp > 77)
```

```
# A tibble: 85 × 6
```

	Ozone	Solar.R	Wind	Temp	Month	Day
	<int>	<int>	<dbl>	<int>	<int>	<int>
1	45	252	14.9	81	5	29
2	115	223	5.7	79	5	30
3	NA	286	8.6	78	6	1
4	NA	186	9.2	84	6	4
5	NA	220	8.6	85	6	5
6	NA	264	14.3	79	6	6
7	29	127	9.7	82	6	7
8	NA	273	6.9	87	6	8
9	71	291	13.8	90	6	9
10	39	323	11.5	87	6	10

```
# i 75 more rows
```

```
# i Use `print(n = ...)` to see more rows
```

Filtrar filas: filter()

tabla %>% filter(...)

dataset

Argumentos de filtrado

```
> aire_tbl
```

```
# A tibble: 153 × 6
```

	Ozone	Solar.R	Wind	Temp	Month	Day
	<int>	<int>	<dbl>	<int>	<int>	<int>
1	41	190	7.4	67	5	1
2	36	118	8	72	5	2
3	12	149	12.6	74	5	3
4	18	313	11.5	62	5	4
5	NA	NA	14.3	56	5	5
6	28	NA	14.9	66	5	6
7	23	299	8.6	65	5	7
8	19	99	13.8	59	5	8
9	8	19	20.1	61	5	9
10	NA	194	8.6	69	5	10

```
# i 143 more rows
```

```
# i Use `print(n = ...)` to see more rows
```

Filtro temperatura por arriba de 77 y mes 5

```
aire_tbl %>% filter(Temp > 77 & Month == 5)
```

```
# A tibble: 2 × 6
```

	Ozone	Solar.R	Wind	Temp	Month	Day
	<int>	<int>	<dbl>	<int>	<int>	<int>
1	45	252	14.9	81	5	29
2	115	223	5.7	79	5	30

Filtrar filas: filter()

mallá %>% filter(...)

dataset

Argumentos de filtrado

```
> aire_tbl
```

```
# A tibble: 153 × 6
```

	Ozone	Solar.R	Wind	Temp	Month	Day
	<int>	<int>	<dbl>	<int>	<int>	<int>
1	41	190	7.4	67	5	1
2	36	118	8	72	5	2
3	12	149	12.6	74	5	3
4	18	313	11.5	62	5	4
5	NA	NA	14.3	56	5	5
6	28	NA	14.9	66	5	6
7	23	299	8.6	65	5	7
8	19	99	13.8	59	5	8
9	8	19	20.1	61	5	9
10	NA	194	8.6	69	5	10

```
# i 143 more rows
```

```
# i Use `print(n = ...)` to see more rows
```

Filtro valores de Ozone distintos de NA

```
aire_tbl %>% filter(!is.na(Ozone))
```

```
# A tibble: 116 × 6
```

	Ozone	Solar.R	Wind	Temp	Month	Day
	<int>	<int>	<dbl>	<int>	<int>	<int>
1	41	190	7.4	67	5	1
2	36	118	8	72	5	2
3	12	149	12.6	74	5	3
4	18	313	11.5	62	5	4
5	28	NA	14.9	66	5	6
6	23	299	8.6	65	5	7
7	19	99	13.8	59	5	8
8	8	19	20.1	61	5	9
9	7	NA	6.9	74	5	11
10	16	256	9.7	69	5	12

```
# i 106 more rows
```

```
# i Use `print(n = ...)` to see more rows
```

Filtrar filas: filter()

mallá %>% filter(...)

dataset

Argumentos de filtrado

```
> aire_tbl
```

```
# A tibble: 153 × 6
```

	Ozone	Solar.R	Wind	Temp	Month	Day
	<int>	<int>	<dbl>	<int>	<int>	<int>
1	41	190	7.4	67	5	1
2	36	118	8	72	5	2
3	12	149	12.6	74	5	3
4	18	313	11.5	62	5	4
5	NA	NA	14.3	56	5	5
6	28	NA	14.9	66	5	6
7	23	299	8.6	65	5	7
8	19	99	13.8	59	5	8
9	8	19	20.1	61	5	9
10	NA	194	8.6	69	5	10

```
# i 143 more rows
```

```
# i Use `print(n = ...)` to see more rows
```

Filtro valores por arriba del promedio de Temp y que sean de los días 2, 7 o 11

```
aire_tbl %>% filter(Temp > mean(Temp) & Day %in% c(2, 7, 11))
```

```
# A tibble: 11 × 6
```

	Ozone	Solar.R	Wind	Temp	Month	Day
	<int>	<int>	<dbl>	<int>	<int>	<int>
1	29	127	9.7	82	6	7
2	NA	259	10.9	93	6	11
3	49	248	9.2	85	7	2
4	77	276	5.1	88	7	7
5	NA	139	8.6	82	7	11
6	9	24	13.8	81	8	2
7	122	255	4	89	8	7
8	NA	137	11.5	86	8	11
9	78	197	5.1	92	9	2
10	20	252	10.9	80	9	7
11	44	236	14.9	81	9	11

mutate()

mutate



Generar variables: mutate()

mall %>% mutate(...)

dataset

Argumentos de creación de variables

```
> aire_tbl
```

```
# A tibble: 153 × 6
```

	Ozone	Solar.R	Wind	Temp	Month	Day
	<int>	<int>	<dbl>	<int>	<int>	<int>
1	41	190	7.4	67	5	1
2	36	118	8	72	5	2
3	12	149	12.6	74	5	3
4	18	313	11.5	62	5	4
5	NA	NA	14.3	56	5	5
6	28	NA	14.9	66	5	6
7	23	299	8.6	65	5	7
8	19	99	13.8	59	5	8
9	8	19	20.1	61	5	9
10	NA	194	8.6	69	5	10

```
# i 143 more rows
```

```
# i Use `print(n = ...)` to see more rows
```

Genero columna con el logaritmo de Ozono

```
aire_tbl %>% mutate(o3_log = log(Ozone))
```

```
# A tibble: 153 × 7
```

	Ozone	Solar.R	Wind	Temp	Month	Day	o3_log
	<int>	<int>	<dbl>	<int>	<int>	<int>	<dbl>
1	41	190	7.4	67	5	1	3.71
2	36	118	8	72	5	2	3.58
3	12	149	12.6	74	5	3	2.48
4	18	313	11.5	62	5	4	2.89
5	NA	NA	14.3	56	5	5	NA
6	28	NA	14.9	66	5	6	3.33
7	23	299	8.6	65	5	7	3.14
8	19	99	13.8	59	5	8	2.94
9	8	19	20.1	61	5	9	2.08
10	NA	194	8.6	69	5	10	NA

```
# i 143 more rows
```

```
# i Use `print(n = ...)` to see more rows
```

Generar variables: mutate()

mall %>% mutate(...)

dataset

Argumentos de creación de variables

```
> aire_tbl
```

```
# A tibble: 153 × 6
```

	Ozone	Solar.R	Wind	Temp	Month	Day
	<int>	<int>	<dbl>	<int>	<int>	<int>
1	41	190	7.4	67	5	1
2	36	118	8	72	5	2
3	12	149	12.6	74	5	3
4	18	313	11.5	62	5	4
5	NA	NA	14.3	56	5	5
6	28	NA	14.9	66	5	6
7	23	299	8.6	65	5	7
8	19	99	13.8	59	5	8
9	8	19	20.1	61	5	9
10	NA	194	8.6	69	5	10

```
# i 143 more rows
```

```
# i Use `print(n = ...)` to see more rows
```

Genero copia de columna Month

```
aire_tbl %>% mutate(mes = Month)
```

```
# A tibble: 153 × 7
```

	Ozone	Solar.R	Wind	Temp	Month	Day	mes
	<int>	<int>	<dbl>	<int>	<int>	<int>	<int>
1	41	190	7.4	67	5	1	5
2	36	118	8	72	5	2	5
3	12	149	12.6	74	5	3	5
4	18	313	11.5	62	5	4	5
5	NA	NA	14.3	56	5	5	5
6	28	NA	14.9	66	5	6	5
7	23	299	8.6	65	5	7	5
8	19	99	13.8	59	5	8	5
9	8	19	20.1	61	5	9	5
10	NA	194	8.6	69	5	10	5

```
# i 143 more rows
```

```
# i Use `print(n = ...)` to see more rows
```

Generar variables: mutate()

mall %>% mutate(...)

dataset

Argumentos de creación de variables

```
> aire_tbl
```

```
# A tibble: 153 × 6
```

	Ozone	Solar.R	Wind	Temp	Month	Day
	<int>	<int>	<dbl>	<int>	<int>	<int>
1	41	190	7.4	67	5	1
2	36	118	8	72	5	2
3	12	149	12.6	74	5	3
4	18	313	11.5	62	5	4
5	NA	NA	14.3	56	5	5
6	28	NA	14.9	66	5	6
7	23	299	8.6	65	5	7
8	19	99	13.8	59	5	8
9	8	19	20.1	61	5	9
10	NA	194	8.6	69	5	10

```
# i 143 more rows
```

```
# i Use `print(n = ...)` to see more rows
```

Genero nueva columna que concatena Month y Day

```
aire_tbl %>% mutate(dia_mes = str_c(Day, Month, sep = "-"))
```

```
# A tibble: 153 × 7
```

	Ozone	Solar.R	Wind	Temp	Month	Day	dia_mes
	<int>	<int>	<dbl>	<int>	<int>	<int>	<chr>
1	41	190	7.4	67	5	1	1-5
2	36	118	8	72	5	2	2-5
3	12	149	12.6	74	5	3	3-5
4	18	313	11.5	62	5	4	4-5
5	NA	NA	14.3	56	5	5	5-5
6	28	NA	14.9	66	5	6	6-5
7	23	299	8.6	65	5	7	7-5
8	19	99	13.8	59	5	8	8-5
9	8	19	20.1	61	5	9	9-5
10	NA	194	8.6	69	5	10	10-5

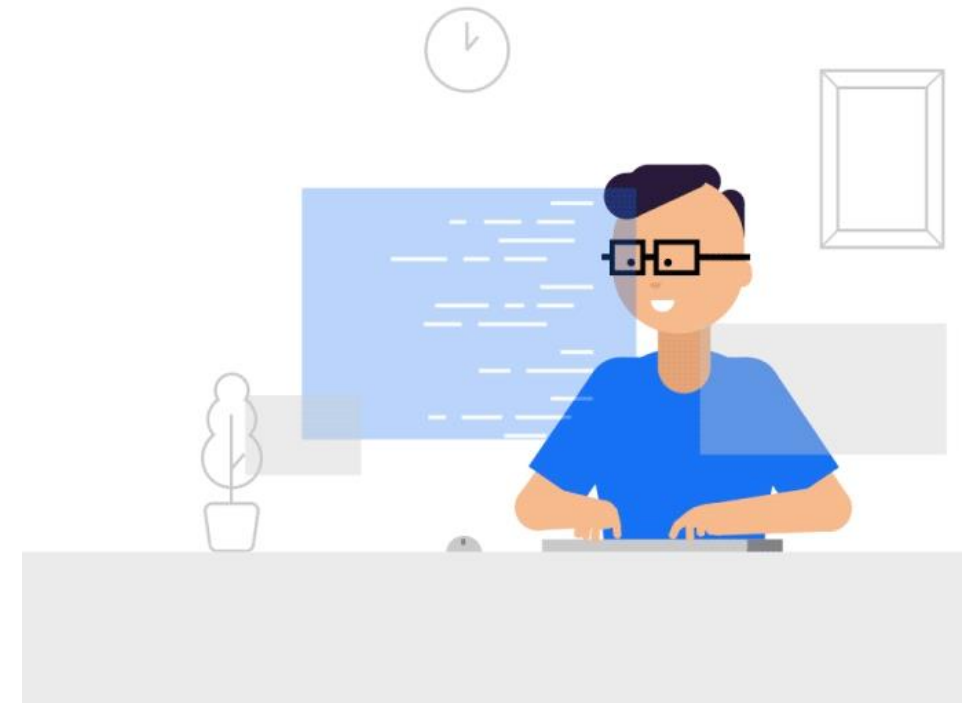
```
# i 143 more rows
```

```
# i Use `print(n = ...)` to see more rows
```



Su turno...

- Queremos una malla de datos que contenga sólo a las columnas `Day`, `Month`, `Ozone` y `Solar.R`
- Queremos aquellos días en los que la concentración de ozono excedió o fue igual al promedio del periodo
- Queremos que los datos sean sólo del mes 7
- Requerimos transformar a logarítmica la variable de ozono
- No guarde el resultado, sólo genere la consulta





```
aire_tbl %>%
```

```
  select(Day, Month, Ozone, Solar.R) %>%
```

```
  filter(Ozone >= mean(Ozone, na.rm = TRUE) & Month == 7) %>%
```

```
  mutate(o3_lag = log(Ozone))
```

```
# A tibble: 26 × 5
```

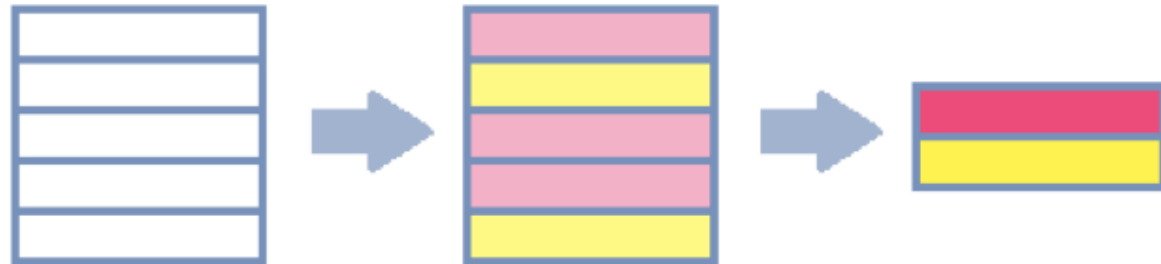
	Day	Month	Ozone	Solar.R	o3_lag
	<int>	<int>	<int>	<int>	<dbl>
1	29	5	45	252	3.81
2	30	5	115	223	4.74
3	9	6	71	291	4.26
4	4	8	78	NA	4.36
5	6	8	66	NA	4.19
6	7	8	122	255	4.80
7	8	8	89	229	4.49
8	9	8	110	207	4.70
9	12	8	44	192	3.78
10	14	8	65	157	4.17

```
# i 16 more rows
```

```
# i Use `print(n = ...)` to see more rows
```

summarise()

summarise





```
aire_tbl %>%
```

```
  summarise(
```

```
    media_ozono = mean(Ozone, na.rm = TRUE),
```

```
    media_temp = mean(Temp, na.rm = TRUE)
```

```
  )
```

summarise()

Obtenemos una medida de
resumen de cada una de las
variables que especifiquemos



```
aire_tbl %>%
```

```
  summarise(
```

```
    media_ozono = mean(Ozone, na.rm = TRUE),
```

```
    media_temp = mean(Temp)
```

```
)
```

```
# A tibble: 1 × 2
```

```
  media_ozono media_temp
```

```
    <dbl>
```

```
    <dbl>
```

```
1
```

```
  42.1
```

```
  77.9
```



Agrupar datos: `group_by()`

```
tabla %>% group_by(...) %>% summarise(...)
```

Malla de datos

Argumentos de
agrupación

Argumentos de
resumen

Agrupar datos... `group_by( )`

mallá

%>%

`group_by(entidad)`

%>%

`summarise(...)`

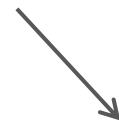
entidad	municipio	población
Aguascalientes	Asientos	51,536
Aguascalientes	Calvillo	58,250
Baja California	Ensenada	443,807
Baja California	Mexicali	1,049,792
Ciudad de México	Azcapotzalco	432,205
Ciudad de México	Benito Juárez	434,153



Aguascalientes	Asientos	51,536
Aguascalientes	Calvillo	58,250

Baja California	Ensenada	443,807
Baja California	Mexicali	1,049,792

Ciudad de México	Azcapotzalco	432,205
Ciudad de México	Benito Juárez	434,153



entidad	pob_pro	pob_to	n
Aguascalientes	54,893.0	109,786	2
Baja California	433,179.0	1,493,599	2
Ciudad de México	740,998.5	866,358	2

`mallá %>%`

`group_by(entidad) %>%`

`summarise(pob_prom = mean(población),`

`pob_tot = sum(población),`

`n = n())`



```
aire_tbl %>%  
  group_by(Month) %>%  
  summarise(  
    media_ozono = mean(Ozone, na.rm = TRUE),  
    media_temp = mean(Temp)  
  )
```

```
# A tibble: 5 × 3  
  Month media_ozono media_temp  
  <int>      <dbl>      <dbl>  
1     5      23.6      65.5  
2     6      29.4      79.1  
3     7      59.1      83.9  
4     8      60.0      84.0  
5     9      31.4      76.9
```



```
aire_tbl %>%  
  group_by(Month) %>%  
  summarise(  
    media_ozono = mean(Ozone, na.rm = TRUE),  
    media_temp = mean(Temp),  
    tot_datos = n(),  
    tot_na_o3 = sum(!is.na(Ozone)),  
    percent_mes_o3 = (tot_na_o3 * 100)/tot_datos  
  )
```

A tibble: 5 × 6

	Month	media_ozono	media_temp	tot_datos	tot_na_o3	percent_mes_o3
	<int>	<dbl>	<dbl>	<int>	<int>	<dbl>
1	5	23.6	65.5	31	26	83.9
2	6	29.4	79.1	30	9	30
3	7	59.1	83.9	31	26	83.9
4	8	60.0	84.0	31	26	83.9
5	9	31.4	76.9	30	29	96.7



distinct()

En ocasiones requerimos
identificar los valores
únicos de una variable

```
aire_tbl %>% distinct(Month)
```

```
# A tibble: 5 × 1
  Month
  <int>
1     5
2     6
3     7
4     8
5     9
```



arrange()

Esta función la utilizamos para ordenar los valores de una variable.

```
aire_tbl %>% arrange(Solar.R)
```

```
# A tibble: 153 × 6
  Ozone Solar.R Wind Temp Month Day
<int>   <int> <dbl> <int> <int> <int>
1     16       7  6.9    74     7    21
2      1       8  9.7    59     5    21
3     23     13  12     67     5    28
4     23     14  9.2    71     9    22
5      8     19 20.1    61     5     9
6     14     20 16.6    63     9    25
7      9     24 13.8    81     8     2
8      9     24 10.9    71     9    14
9      4     25  9.7    61     5    23
10     13     27 10.3    76     9    18
# i 143 more rows
# i Use `print(n = ...)` to see more rows
```



arrange()

Esta función la utilizamos para ordenar los valores de una variable.

```
aire_tbl %>%  
  distinct(Month) %>%  
  arrange(desc(Month))
```

```
# A tibble: 5 × 1  
  Month  
  <int>  
1     9  
2     8  
3     7  
4     6  
5     5
```



relocate()

Esta función la utilizamos para mover una columna a otra posición.

```
# A tibble: 153 × 6
```

	Ozone	Solar.R	Wind	Temp	Month	Day
	<int>	<int>	<dbl>	<int>	<int>	<int>
1	41	190	7.4	67	5	1
2	36	118	8	72	5	2
3	12	149	12.6	74	5	3
4	18	313	11.5	62	5	4
5	NA	NA	14.3	56	5	5
6	28	NA	14.9	66	5	6
7	23	299	8.6	65	5	7
8	19	99	13.8	59	5	8
9	8	19	20.1	61	5	9
10	NA	194	8.6	69	5	10

```
# i 143 more rows
```

```
# i Use `print(n = ...)` to see more rows
```

```
aire_tbl %>%
```

```
  relocate(Day, Month, .before = Ozone)
```

```
# A tibble: 153 × 6
```

	Day	Month	Ozone	Solar.R	Wind	Temp
	<int>	<int>	<int>	<int>	<dbl>	<int>
1	1	5	41	190	7.4	67
2	2	5	36	118	8	72
3	3	5	12	149	12.6	74
4	4	5	18	313	11.5	62
5	5	5	NA	NA	14.3	56
6	6	5	28	NA	14.9	66
7	7	5	23	299	8.6	65
8	8	5	19	99	13.8	59
9	9	5	8	19	20.1	61
10	10	5	NA	194	8.6	69

```
# i 143 more rows
```

```
# i Use `print(n = ...)` to see more rows
```



relocate()

Esta función la utilizamos para mover una columna a otra posición.

```
# A tibble: 153 × 6
  Ozone Solar.R Wind Temp Month Day
  <int>   <int> <dbl> <int> <int> <int>
1    41    190  7.4    67     5     1
2    36    118   8     72     5     2
3    12    149 12.6    74     5     3
4    18    313 11.5    62     5     4
5    NA     NA 14.3    56     5     5
6    28     NA 14.9    66     5     6
7    23    299  8.6    65     5     7
8    19     99 13.8    59     5     8
9     8     19 20.1    61     5     9
10    NA    194  8.6    69     5    10
# i 143 more rows
# i Use `print(n = ...)` to see more rows
```

```
aire_tbl %>%
```

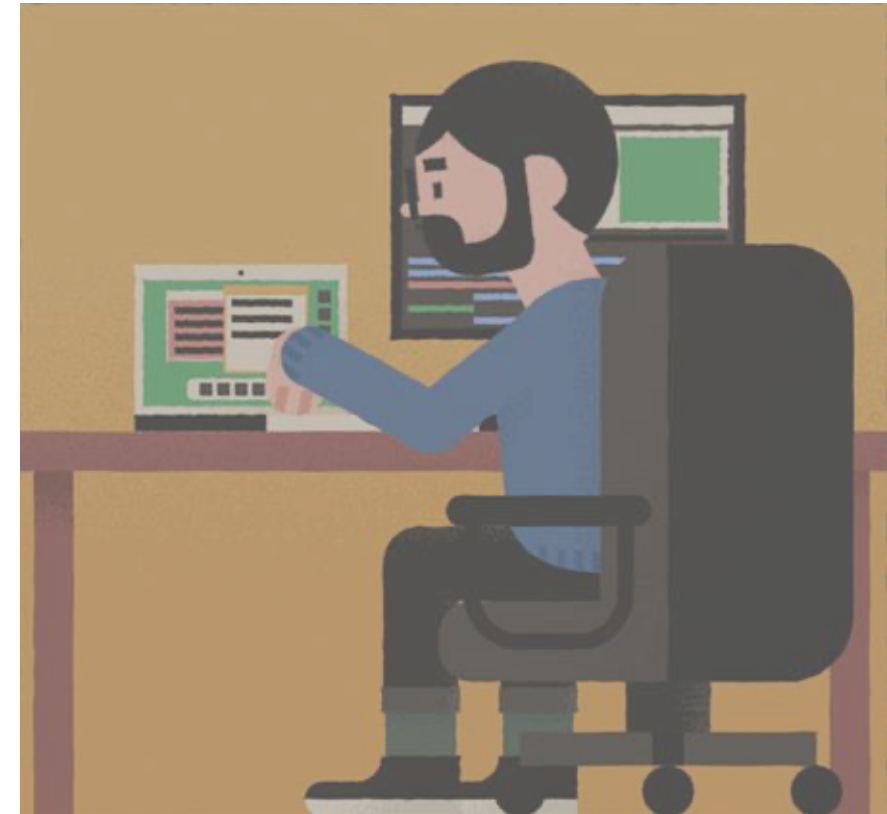
```
  relocate(Temp, .after = Ozone)
```

```
# A tibble: 153 × 6
  Ozone Temp Solar.R Wind Month Day
  <int> <int>   <int> <dbl> <int> <int>
1    41    67    190  7.4     5     1
2    36    72    118   8     5     2
3    12    74    149 12.6     5     3
4    18    62    313 11.5     5     4
5    NA    56     NA 14.3     5     5
6    28    66     NA 14.9     5     6
7    23    65    299  8.6     5     7
8    19    59     99 13.8     5     8
9     8    61     19 20.1     5     9
10    NA    69    194  8.6     5    10
# i 143 more rows
# i Use `print(n = ...)` to see more rows
```



Su turno...

- Importe a su sesión el archivo `red_manual_particulas_susp.csv` y nombre al objeto como `redma`
- ¿qué parámetros se miden?
- ¿cuál es la fecha más antigua y más reciente de la malla?
- ¿qué estación o estaciones tienen la concentración más baja de PM_{10} ?





¿qué parámetros se miden?

```
aire_tbl %>%  
  distinct(id_parameter)
```

```
# A tibble: 3 × 1  
  id_parameter  
  <chr>  
1 PM10  
2 PST  
3 PM25
```



Fecha más antigua de la malla

```
summary(aire_tbl$id_parameter)
```

```
> summary(redma$date)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
"1989-01-02"	"1997-03-27"	"2006-05-29"	"2006-06-23"	"2015-06-05"	"2024-12-26"



Concentración más baja de PM₁₀

```
aire_tbl %>%  
  filter(id_parameter == "PM10") %>%  
  arrange(value)
```

```
# A tibble: 19,927 × 5
```

	date	id_station	value	id_parameter	unit
	<date>	<chr>	<dbl>	<chr>	<dbl>
1	2018-09-17	LOM	0	PM10	2
2	2021-06-27	UIZ	4	PM10	2
3	1997-04-20	PED	5	PM10	2
4	2014-10-20	PED	5	PM10	2
5	2021-06-27	PED	5	PM10	2
6	2016-10-09	LPR	6	PM10	2
7	2017-09-10	LPR	6	PM10	2
8	2021-06-27	MER	6	PM10	2
9	2021-06-27	NEZ	6	PM10	2
10	2024-09-27	PED	6	PM10	2

```
# i 19,917 more rows
```

```
# i Use `print(n = ...)` to see more rows
```



```
pm25_2024 <-  
  redma %>%  
  filter(id_parameter == "PM25" & date >= "2024-01-01") %>%  
  select(date, id_station, value) %>%  
  rename(site = id_station, pm25 = value) %>%  
  print()
```

```
# A tibble: 427 × 3
```

	date	site	pm25
	<date>	<chr>	<dbl>
1	2024-01-01	COY	NA
2	2024-01-01	MER	75
3	2024-01-01	PED	35
4	2024-01-01	SAG	79
5	2024-01-01	TLA	46
6	2024-01-01	UIZ	83
7	2024-01-01	XAL	113
8	2024-01-07	COY	NA
9	2024-01-07	MER	15
10	2024-01-07	PED	8

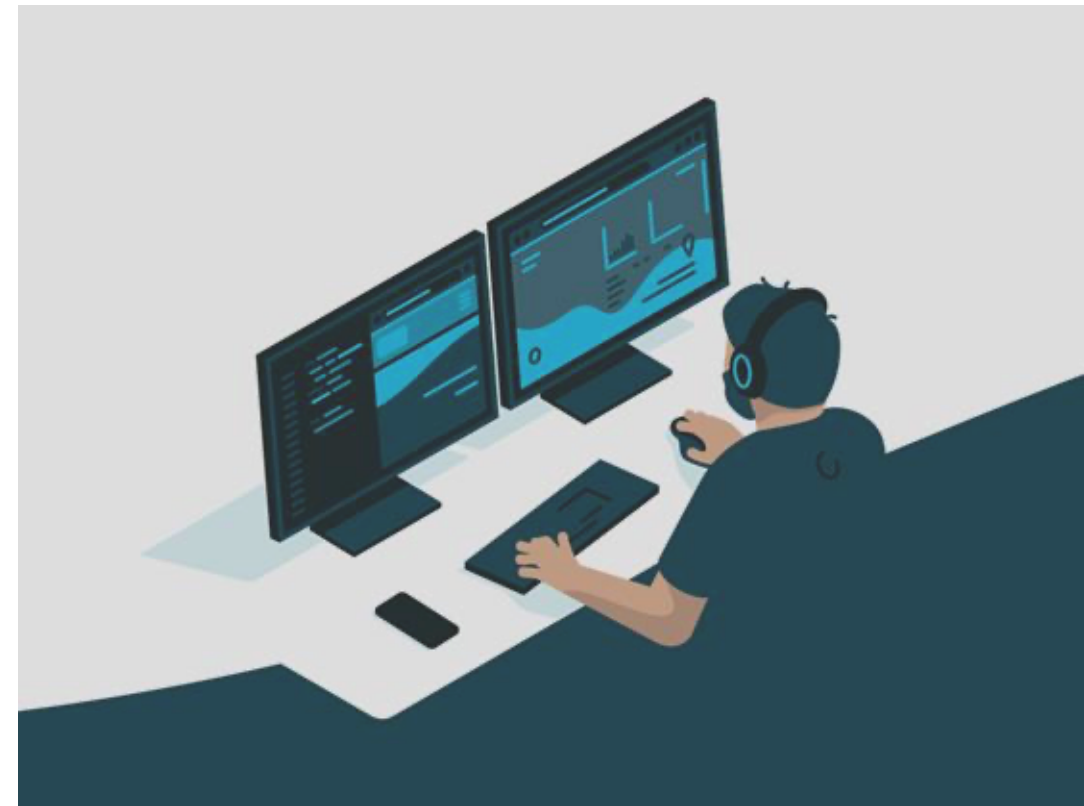
```
# i 417 more rows
```

```
# i Use `print(n = ...)` to
```



Malla pm25_2024

- Transforme la variable `pm25` a logarítmica, nombre la nueva variable como `pm25_log`
- Revise nuevamente su gráfico



Editar componentes del gráfico

ggplot2

theme elements reference

Set minimal as the baseline theme:

```
theme_minimal() +  
theme(theme.element = element_type())
```

Use `element_blank()` to remove an element

Axis titles, text, ticks, and lines can be specified per axis using theme inheritance by putting `.x/.y` at the end of the theme element.

`axis.line.y = element_line()`

`axis.title.y = element_text()`

`panel.grid.major = element_line()`

`panel.grid.minor = element_line()`

`axis.text.y`

`axis.text = element_text()`

`axis.text.x`

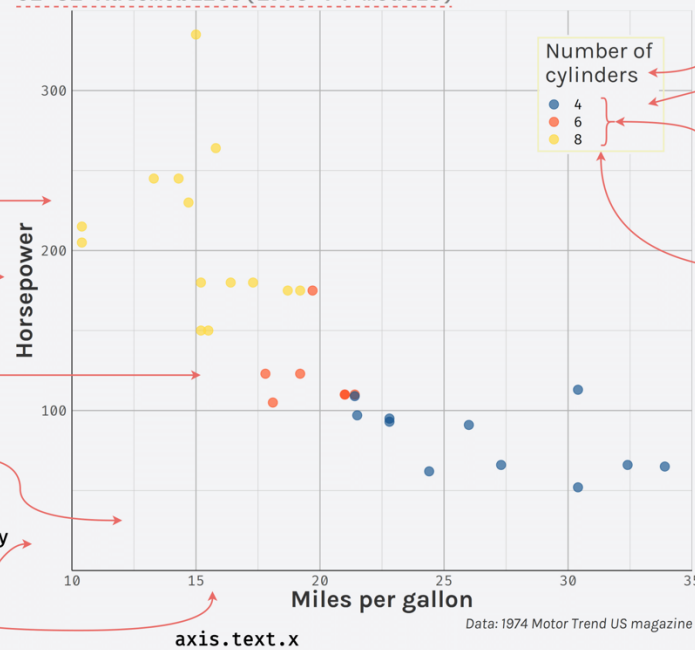
`text = element_text()` ← modifications will be applied to all text elements

```
plot.title.position = "plot"  
plot.caption.position = "plot"  
plot.title = element_text()  
plot.subtitle = element_text()
```

"} "plot" means that they will be aligned to the entire plot (instead of the panel)

`plot.margin = margin(25, 25, 25, 25)`

Miles per Gallon & Horsepower
of 32 Automobiles(1973-74 models)



`legend.title = element_text()`

`legend.background = element_rect()`

`legend.text = element_text()`

`legend.position = c(.85, .85) / "none" / "left" / "right" / "bottom" / "top"`

`plot.background = element_rect()`

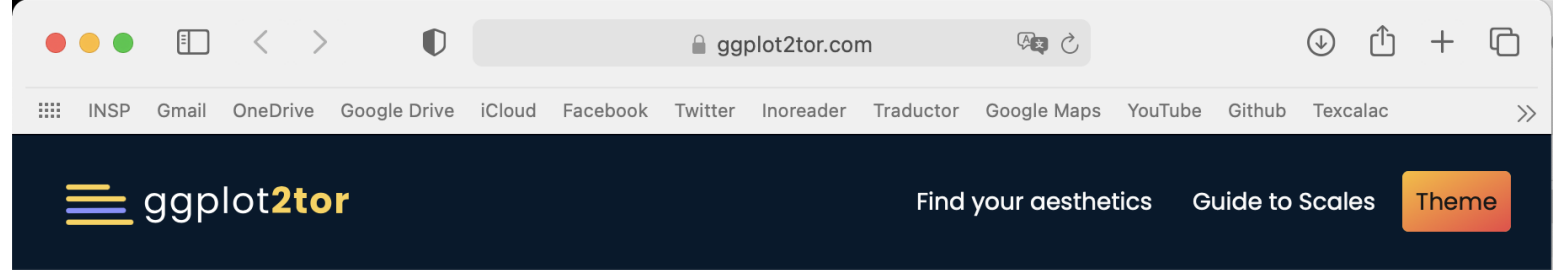
`plot.caption = element_text()`

isabella-b

Full list of elements at ggplot2.tidyverse.org/reference/theme/



<https://ggplot2tor.com/theme/>



A complete guide on how to modify the ggplot2 theme



You can use the ggplot2 theme to change the appearance of your visualization. However, it's hard to know all the elements by heart. Use this app as a quick reference to create your own theme. Just navigate through the elements, click to open the sub-navigation and copy and paste each argument straight into our code.

Hover over the theme elements, click to open the sub-navigation and copy the code.

plot ▾

panel ▾

axis ▾

strip ▾

legend ▾